



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



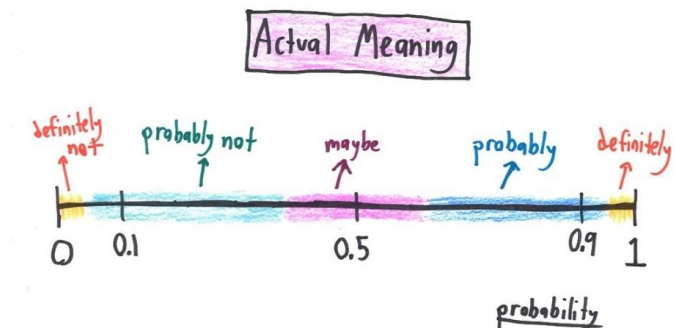


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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Week 5 - Lecture No 07 – 15th September 2025



Announcements!

1. Lectures up to No 06 posted
2. Additional lecture - September 20, 2025 (Saturday, same time & place) in-lieu of the class missed on Wednesday August 20, 2025
3. An added lecture will be posted in a asynchronous mode in lieu of September 08, 2025 class
4. Quiz No 02 – September 17/19/20/21, 2025

Agenda for today

1. Complete Unit 2 - Conditional Probability and Bayes' Rule
2. Unit 3 - Counting

Independence of a collection of events

- **Intuitively:** Information on some of the events does not change probabilities related to the remaining events.

→ $A_1, A_2, \dots, A_n$ are **independent** $\Rightarrow P(A_3 \cap A_4^c) = P(A_3 \cap A_4^c \mid A_1 \cup (A_2 \cap A_5^c))$

$$\rightarrow P(A_3) = P(A_3 \mid A_1 \cap A_2) = P(A_3 \mid A_1 \cap A_2^c) = P(A_3 \mid A_1^c \cap A_2)$$

Definition: Events $A_1, A_2, \dots, A_n$ are called **independent** if:

$$P(A_i \cap A_j \cap \dots \cap A_m) = P(A_i) P(A_j) \cdots P(A_m) \quad \text{for any distinct indices } i, j, \dots, m$$

$n = 3$:

$$\left. \begin{aligned} P(A_1 \cap A_2) &= P(A_1) \cdot P(A_2) \\ P(A_1 \cap A_3) &= P(A_1) \cdot P(A_3) \\ P(A_2 \cap A_3) &= P(A_2) \cdot P(A_3) \end{aligned} \right\} \text{Pairwise independence}$$

Does pairwise independence **imply** $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2)P(A_3)$?

Independence vs pairwise independence

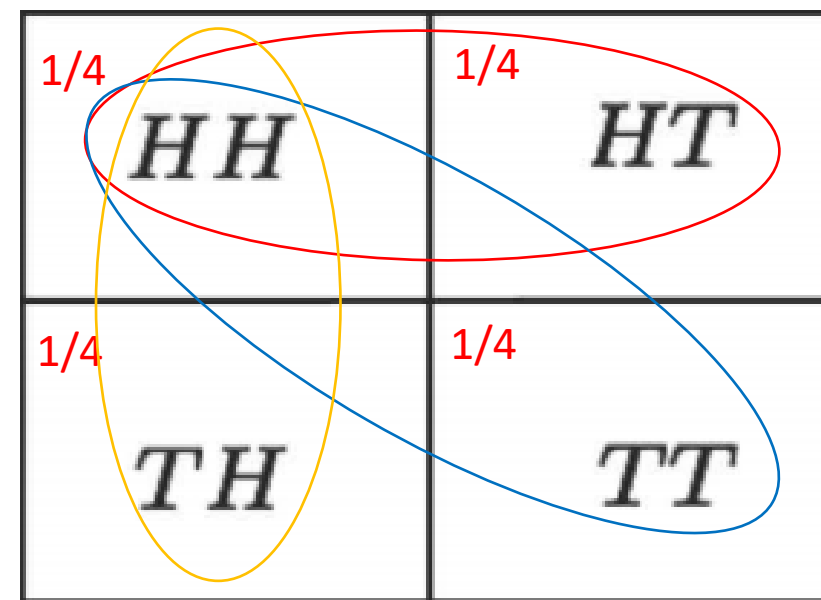
- Two independent fair coin tosses

➤ H_1 : First toss is H

➤ H_2 : Second toss is H

$$P(H_1) = P(H_2) = \frac{1}{2}$$

- C : the two tosses had the same result



$$P(H_1 \cap C) = P(H_1 \cap H_2) = 1/4 \quad P(H_1) * P(C) = \frac{1}{2} \times \frac{1}{2} = 1/4$$

H_1 & C are pairwise independent & so are H_2 & C , & we already know about H_1 & H_2

$$P(H_1 \cap H_2 \cap C) = P(\{HH\}) = 1/4 \quad P(H_1) * P(H_2) * P(C) = 1/8$$

H_1 , H_2 & C are collectively not independent

Intuitively, $P(C|H_1) = P(H_2|H_1) = P(H_2) = \frac{1}{2} = P(C)$

$$P(C|H_1 \cap H_2) = 1 \neq P(C)$$

H_1 , H_2 and C are pairwise independent, but not independent.

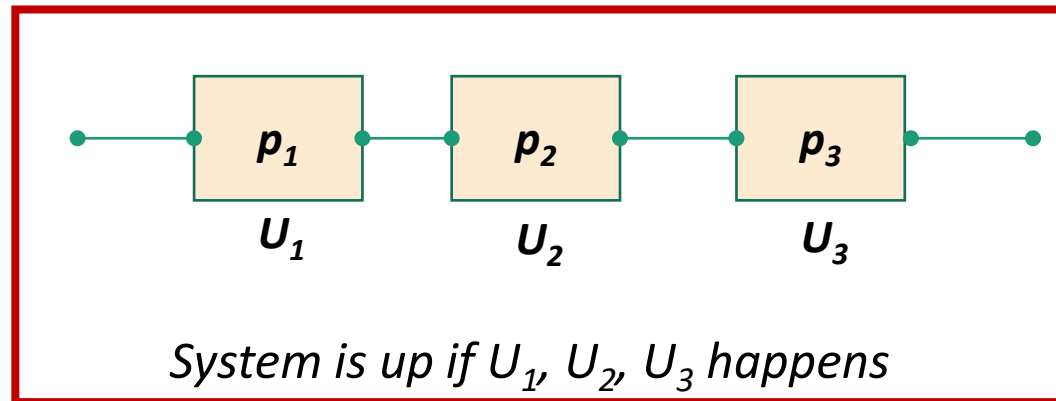
Application of Independence: Reliability (systems in Series)

- p_j : probability that unit j is “up”

Unit i is serviceable w.p p_i & denoted by U_i

Unit i is unserviceable w.p $(1 - p_i)$ & denoted by F_i

➤ $U_{i's}$ are independent, & therefore $F_{i's}$ are also independent



➔
$$\begin{aligned} P(\text{System is up}) &= P(U_1 \cap U_2 \cap U_3) = P(U_1) \cdot P(U_2) \cdot P(U_3) \\ &= p_1 \cdot p_2 \cdot p_3 \end{aligned}$$

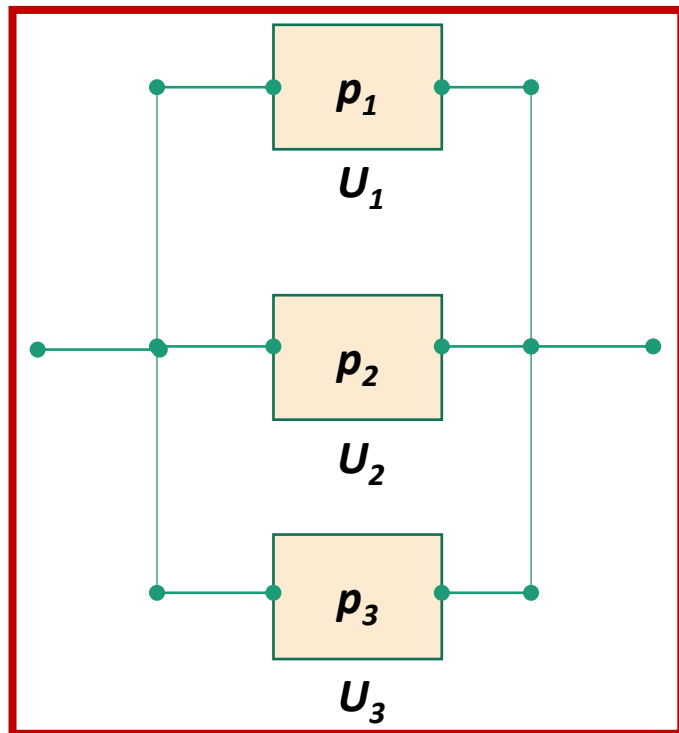
Application of Independence: Reliability (systems in Parallel)

- p_j : probability that unit j is “up”

Unit i is serviceable w.p p_i & denoted by U_i

Unit i is unserviceable w.p $(1 - p_i)$ & denoted by F_i

➤ U_i s are independent, & therefore F_i s are also independent



$$\begin{aligned} \Rightarrow P(\text{System is up}) &= P(U_1 \cup U_2 \cup U_3) \\ \Rightarrow &= 1 - P(U_1 \cup U_2 \cup U_3)^c \\ \Rightarrow &= 1 - P(U_1^c \cap U_2^c \cap U_3^c) \\ \Rightarrow &= 1 - P(F_1 \cap F_2 \cap F_3) \\ \Rightarrow &= 1 - P(F_1) \cdot P(F_2) \cdot P(F_3) \\ \Rightarrow &= 1 - (1 - p_1) \cdot (1 - p_2) \cdot (1 - p_3) \end{aligned}$$

System is up if Path exists between A & B

Practice Problem 1 –

In a specified 6-AM-to-6-AM 24-hour period, a student wakes up at time t_1 and goes to sleep at some later time t_2 . Assume that 6AM at the start of cycle is 0, 7AM is 1 and so on, so that next day 6AM is time $t = 24$. Therefore, the sample space is given by (assume equally likely probability for wake up and sleep time): –

$$\Omega = \{(t_1, t_2) | t_1 \in [0, 24), t_2 \in (0, 24], \text{ and } t_2 > t_1\}$$

Calculate the probability by specifying the set A and sketch the region on the plane corresponding to the event that the “*student is asleep at noon*”.